

Tutorial 7: Inter Process Communication*Nanyang Technological University*

- Q1 (a)** When is there a need to use IPC?
- (b)** Comparing the three IPC methods: Anonymous Pipe, Named Pipe and Sockets
- what are the common features used in these methods?
 - state the main feature of each of the methods
- Q2** Explain the meaning of Blocking and Non-blocking and discuss how they are most suitably used between a sender and receiver.
- Q3** What is the limitation of named pipe for IPC?
- Q4** Suppose you have two processes, Process A and Process B, that use two named pipes, P1 and P2, for communication. Process A is responsible for writing data to P1 and reading data from P2, and Process B reads from P1 and writes to P2. Process A reads and writes the data at the same time in a thread. Process B does the same thread. What is the potential problem? How do you overcome it?
- Q5** Create a named pipe named “my_pipe” in your current directory. Write a script in Bash named “producer.sh” that continually prompts the user for input and writes that input to “my_pipe” until the user enters “quit”. Similarly, create another script named “consumer.sh” that reads from “my_pipe” and prints the received messages to the terminal. The consumer script should keep reading from the pipe until it encounters an end-of-file condition or receives the message “quit”. After creating both scripts, demonstrate the functionality by running the consumer script in one terminal window and the producer script in another. Test the communication by entering messages in the producer window and observing the output in the consumer window. End the demonstration by entering “quit” in the producer window.